

 **TARGET CAT 2025** 



MBA FASTRACK

QUANT : Arithmetic

Time Speed Distance

Part II

Lecture No.- 08

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Recap of Previous Lecture :

$$S = \frac{d}{t}$$

$$t = \frac{d}{s}$$

$$S \times t = d$$

$$1 \text{ km/h} = \frac{5}{18} \text{ m/s}$$

$$1 \text{ m/s} = \frac{18}{5} \text{ km/h}$$

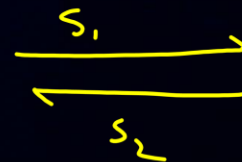
$$1 \text{ min} = \frac{1}{60} \text{ h}$$

$$\text{Avg. Speed} = \frac{T \cdot \text{dist}}{T \cdot \text{time}}$$

→ Generalised

$$\text{Avg. Speed} = \frac{2 S_1 S_2}{S_1 + S_2}$$

→ dist is same





Recap of Previous Lecture :

t constant

$$s \propto d$$

$$\frac{s_1}{s_2} = \frac{d_1}{d_2}$$

d constant

$$s \propto \frac{1}{t}$$

$$\frac{s_1}{s_2} = \frac{t_2}{t_1}$$

s constant

$d \propto t$

$$\frac{d_1}{d_2} = \frac{t_1}{t_2}$$

Relative speed

Same

$$s_r = s_1 - s_2$$

Oppo.

$$s_r = s_1 + s_2$$

TOPICS *to be covered*



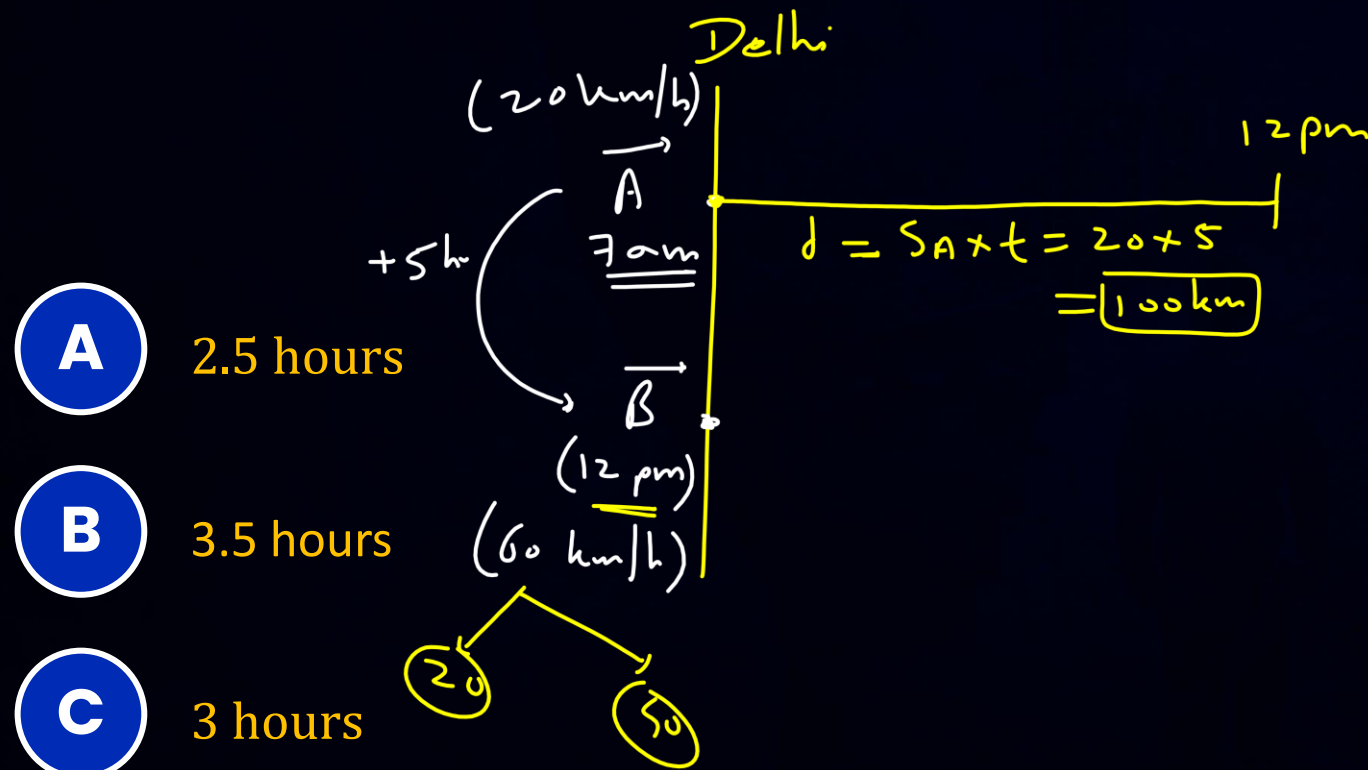
- 1) ~~Concept~~ & Numerical based on 'Problems on Train' ✓
- 2) ~~Concept~~ & Numerical based on 'Boats & Stream' ✓
- 3) Concept & Numerical based on 'Escalator'
- 4) Problems for Practice & CAT PYQs

QUESTION-1

$$d = s \times t$$



#Q. Car A leaves Delhi at a certain time, after 5 hours Car B leaves Delhi in the same direction as of A. Speed of Car A is 20 km/hr and Speed of Car B is 60 km/hr. How much time Car B will take to overtake Car A?



$$\begin{aligned} s_R &= s_B - s_A \\ &= 60 - 20 \\ &= 40 \text{ km/h} \\ d &= 100 \text{ km} \end{aligned}$$

$$t = \frac{d}{s_R} = \frac{100}{40} = 2.5 \text{ h}$$

- A** 2.5 hours
- B** 3.5 hours
- C** 3 hours
- D** 2 hours

QUESTION- 2

$$1 \text{ km/h} = \frac{5}{18} \text{ m/s}$$



#Q. Ravi is driving at a speed of 40 km/h on a road. Vijay is 54 meters behind Ravi and driving in the same direction as Ravi. Ashok is driving along the same road from the opposite direction at a speed of 50 km/h and is 225 meters away from Ravi. The speed, in km/h, at which Vijay should drive so that all the three cross each other at the same time, is

[CAT 2023 : Slot 2]

$$S_R = R + A = 40 + 50$$

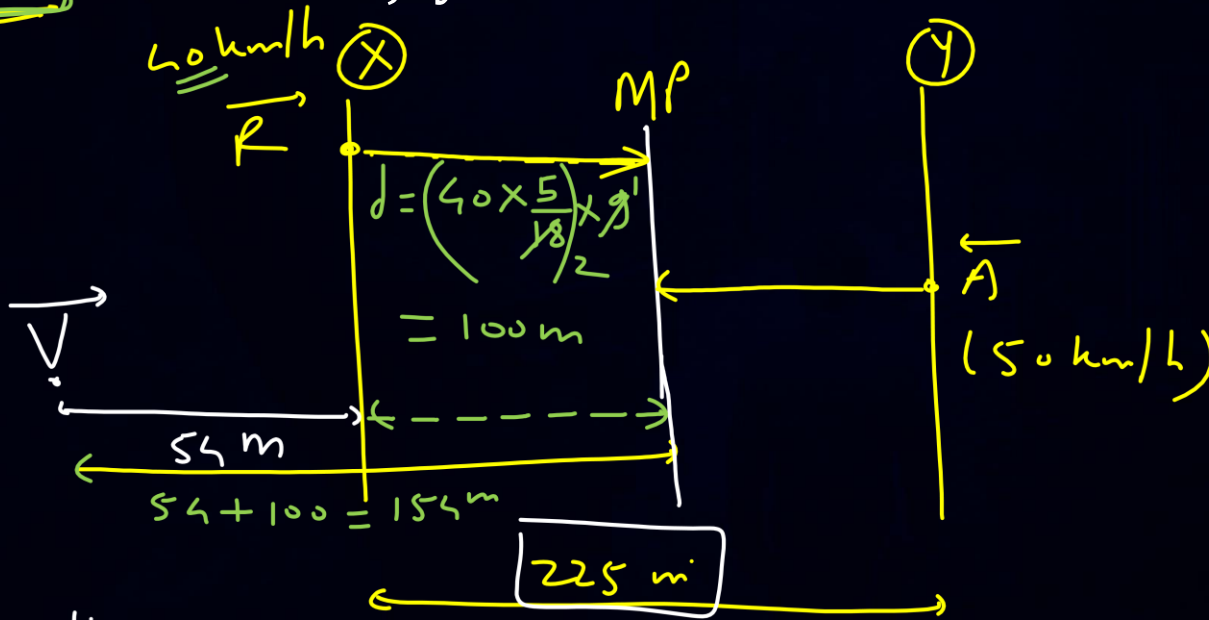
$$= 90 \text{ km/h}$$

$$= 90 \times \frac{5}{18} \text{ m/s}$$

$$= \boxed{25 \text{ m/s}}$$

$$d = 225 \text{ m}$$

$$t = \frac{d}{S_R} = \frac{225}{25} = 9 \text{ sec}$$



$$t = 9 \text{ sec}$$

$$d = 154 \text{ m}$$

$$S = \frac{d}{t} = \frac{154}{9} \text{ m/s} = \frac{154}{9} \times \frac{18}{5} \text{ km/h} = \frac{308}{5} = 61.6$$

A 58.8

B 64.4

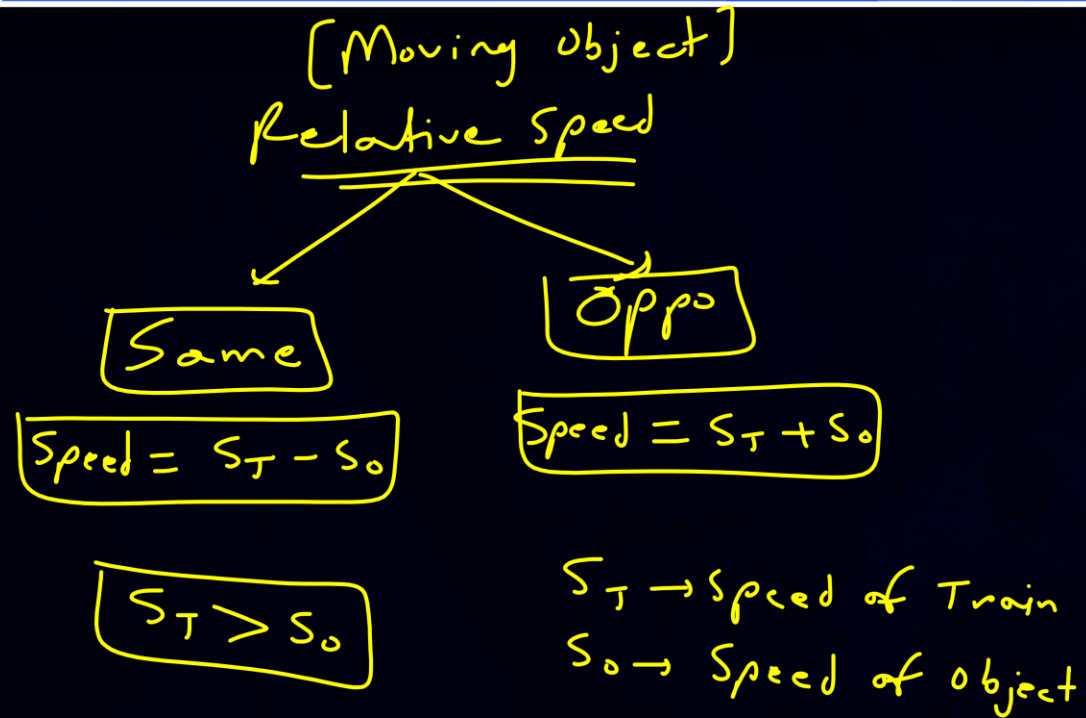
C 67.2

D 61.6



Problems on Train :

$$d = s \times t$$



Stationary Object

$$\text{Speed} = s_T$$



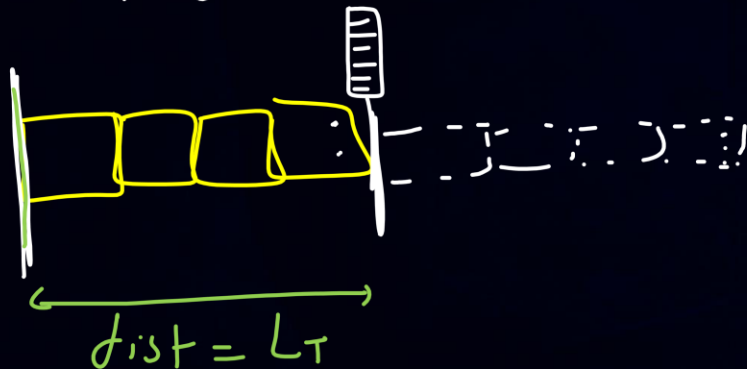
Problems on Train :

$$d = s \times t$$



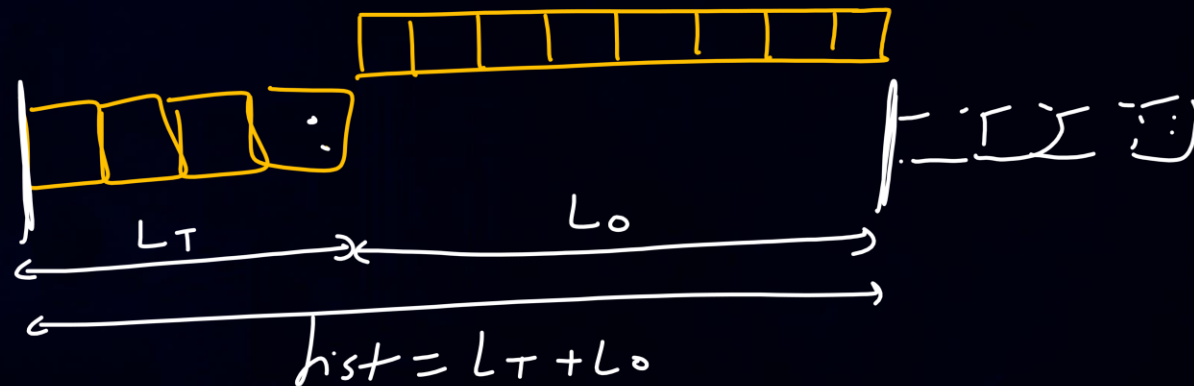
Without Length

[Man/dog/Pole/Tree etc]



With Length

[Another Train/Bridge/Platform/Tunnel etc]



$L_T \rightarrow$ Length of Train
 $L_o \rightarrow$ Length of Object



Problems on Train :

$$S \times t = d$$



1. Train crossing Stationary object without Length

$$\underline{\underline{S_t}} \times \underline{\underline{t}} = L_t$$

2. Train crossing Stationary object with Length

$$\underline{\underline{S_t}} \times \underline{\underline{t}} = L_t + L_o$$

3. Train crossing Moving object without Length

(i) Opposite Direction

$$(\underline{\underline{S_t + S_0}}) \times \underline{\underline{t}} = L_t$$

(ii) Same Direction

$$(\underline{\underline{S_t - S_0}}) \times \underline{\underline{t}} = L_t$$

4. Train crossing Moving object with Length

(i) Opposite Direction

$$(\underline{\underline{S_t + S_0}}) \times \underline{\underline{t}} = (L_t + L_o)$$

(ii) Same Direction

$$(\underline{\underline{S_t - S_0}}) \times \underline{\underline{t}} = (L_t + L_o)$$

QUESTION- 3

$$S_T = 54 \text{ km/hr} = 54 \times \frac{5}{18} \text{ m/s} = 15 \text{ m/s}$$



#Q. A train passes a station platform in 36 seconds and a man standing on the platform in 20 seconds. If the speed of the train is 54 km/hr, what is the length of the platform?

(with Length)

P/F

$$t = 36 \text{ sec}$$

$$L_0 = ?$$

$$S_T \times t = L_T + L_0$$

$$15 \times 36 = 300 + L_0$$

$$540 = 300 + L_0$$

$$240 = L_0$$

(without length) $S \times t = d$

Standing Man

$$t = 20 \text{ sec}$$

$$S_T \times t = L_T$$

$$15 \times 20 = L_T$$

$$300 = L_T$$

A 120 m

B 240 m

C 300 m

D None of these

QUESTION- 4

$$S_T = 48 \text{ km/h} = 48 \times \frac{5}{18} \text{ m/s} = \frac{40}{3} \text{ m/s}$$



#Q. A train travelling at 48 kmph completely crosses another train having half its length and travelling in opposite direction at 42 kmph, in 12 seconds. It also passes a railway platform in 45 seconds. The length of the platform is

(with length)
Another Train

$$(S_T + S_0) \times t = L_T + L_0$$

$$25 \times 12 = 2x + x$$

$$300 = 3x$$

$$100 = x$$

P/F (with length)

$$t = 45 \text{ sec}$$

$$S_T \times t = L_T + L_0$$

$$\frac{40}{3} \times \frac{15}{1} = 200 + L_0$$

$$600 = 200 + L_0$$

$$400 = L_0$$



$$t = 12 \text{ sec}$$

$$\begin{aligned} (S_T + S_0) &= 48 + 42 = 90 \text{ km/h} \\ &= 90 \times \frac{5}{18} \text{ m/s} \\ &= 25 \text{ m/s} \end{aligned}$$

A 400 m

B 450 m

C 560 m

D 600 m

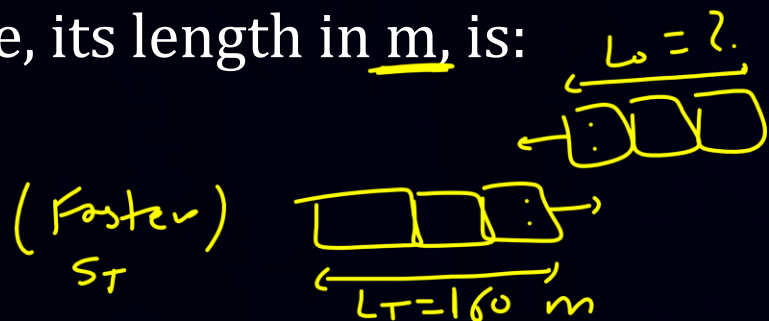
QUESTION- 5 $(S_T + S_o) = (48 + 42) = 90 \text{ km/hr} = 90 \times \frac{5}{18} \text{ m/s} = 25 \text{ m/s}$

$$S \times t = d$$



#Q. Two Trains cross each other in 14 seconds when running in opposite directions along parallel tracks. The faster train is 160 m long and crosses a lamp post in 12 seconds. If the speed of the other train is 6 km/hr less than the faster one, its length in m, is:

[CAT 2021 : Slot 1]



$$\begin{aligned} (S_o &= S_T - 6) \\ &= 48 - 6 \\ &= 42 \text{ km/hr} \end{aligned}$$

Lamp Post (without length) $t = 12 \text{ sec}$

$$S_T \times t = L_T$$

$$S_T \times 12 = 160$$

$$S_T = \frac{160}{12} = \frac{40}{3} \text{ m/s}$$

$$\begin{aligned} \therefore S_T &= \frac{40}{3} \times \frac{18}{5} \text{ km/hr} \\ &= 48 \text{ km/hr} \end{aligned}$$

$$(S_T + S_o) \cdot t = L_T + L_o$$

$$25 \times 14 = 160 + L_o$$

$$350 = 160 + L_o$$

$$190 = L_o$$

A 184

B 192

C 190

D 180

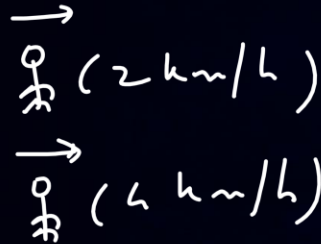
QUESTION- 6

$$S_T \times t = L_T$$

$$S \times t = d$$



#Q. Two persons are walking beside a railway track at respective speeds of 2 & 4 km per hour in the same direction. A train came from behind them and crossed them in 90 & 100 seconds, respectively. The time in seconds, taken by the train to cross an electric post is nearest to:



(Same dir)
Ist Person

$$S_0 = 2 \text{ km/h}$$

$$t = 90 \text{ sec}$$

$$(S_T - S_0) \times t = L_T$$

$$(S_T - 2) \cdot \frac{5}{18} \times 90 = L_T$$

$$(S_T - 2) \cdot \frac{5}{18} \cdot 90 = (S_T - 4) \cdot \frac{5}{18} \cdot 100$$

$$9(S_T - 2) = 10(S_T - 4)$$

$$9S_T - 18 = 10S_T - 40$$

$$22 = S_T$$

$$\text{km/h}$$

[CAT 2020 : Slot 1]

IInd person (Same dir)

$$S_0 = 4 \text{ km/h}$$

$$t = 100 \text{ sec}$$

$$(S_T - S_0) \times t = L_T$$

$$(S_T - 4) \cdot \frac{5}{18} \cdot 100 = L_T$$

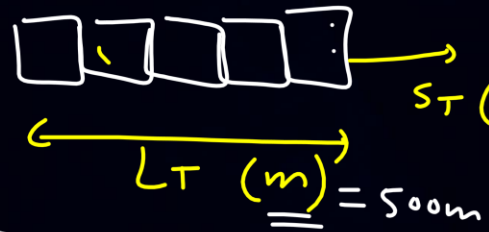
$$\frac{18 \times 5}{18 \times 100} = L_T$$

$$500 = L_T$$

$$\text{m}$$

A

78



B

82

Ele. Post

C

87

$$22 \times \frac{5}{18} \times t = \frac{100}{1000}$$

D

75

$$t = \frac{900}{11} = 81.81$$

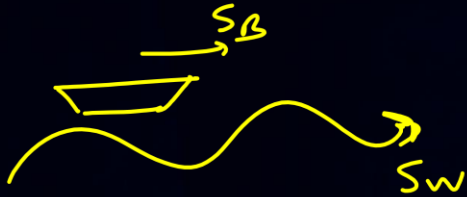
$$\approx 82$$



Boats & Stream :



Down stream / with the flow / Along the flow



$$\text{Speed} = S_B + S_w$$

$S_B \rightarrow$ Speed of Boat in still water

$S_w \rightarrow$ Speed of water / current / stream / flow / river



Boats & Stream :



Upstream / Against the flow



$$\text{Speed} = S_B - S_w$$

$$S_{DS} = S_B + S_w$$

$$S_{up} = S_B - S_w$$

QUESTION- 7



$$S_B > S_w$$

$$S_B = x$$

$$S_w = 3$$

#Q. A boat covers a certain distance downstream in 1 hour, while it comes back in 1.5 hours. If the speed of the stream be 3 kmph, what is the speed of the boat in still water?

$$\frac{S_1}{S_2} = \frac{t_2}{t_1}$$

DS	UP
$t_{DS} = 1 \text{ hr}$	$t_{UP} = 1.5 \text{ hr}$
$S_{DS} = S_B + S_w$	$S_{UP} = S_B - S_w$

← distance is same (constant) →

$$\frac{DS}{UP}$$

$$S \propto \frac{1}{t}$$

$$\frac{S_{DS}}{S_{UP}} = \frac{t_{UP}}{t_{DS}}$$

$$\frac{S_B + S_w}{S_B - S_w} = \frac{1.5}{1}$$

$$\frac{x+3}{x-3} = 1.5$$

$$x+3 = 1.5x - 4.5$$

$$7.5 = 0.5x$$

$$\frac{7.5}{0.5} = x$$

$$\frac{7.5}{0.5} = x$$

$$15 = x$$

- A 12 kmph
- B 13 kmph
- C 14 kmph
- D 15 kmph

QUESTION- 8

$$S_w = 3 \text{ km/h} \quad S_B = v = 9 \text{ km/h}$$

$$t = \frac{d}{S}$$



#Q. A motorboat takes twice as long to go upstream as to return downstream over same distance. The current speed is 3 km/h. What is the time taken by the boat, if total two-way distance is 72 km?

$$t_{up} = 2 t_{ds}$$

$$\frac{t_{up}}{t_{ds}} = \frac{2}{1}$$

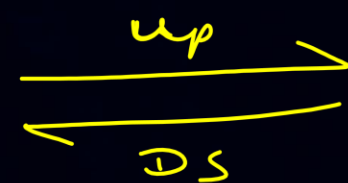
$$\frac{S_{ds}}{S_{up}} = \frac{t_{up}}{t_{ds}} = \frac{2}{1}$$

$$\frac{S_B + S_w}{S_B - S_w} = \frac{2}{1}$$

$$\frac{x+3}{x-3} = \frac{2}{1}$$

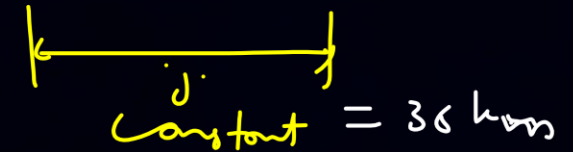
$$x+3 = 2x-6$$

$$9 = x$$



$$2d = 72$$

$$d = 36$$



A 12 hours

B 9 hours

C 15 hours

D 8 hours

$$t_{up} + t_{ds} = T \cdot \text{time}$$

$$\frac{36}{S_B - S_w} + \frac{36}{S_B + S_w} = T \cdot \text{time}$$

$$\frac{36}{6} + \frac{36}{12} = 6 + 3 = 9 \text{ hr} = T \cdot \text{time}$$

QUESTION- 9

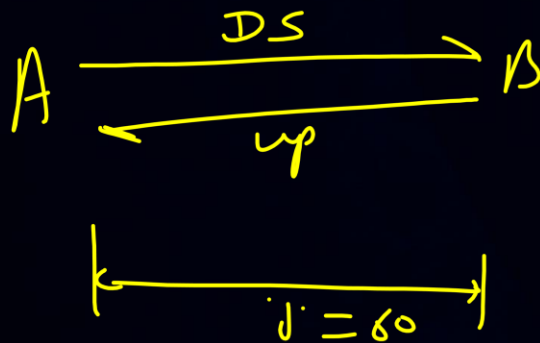
$$S_B = 2.$$

$$t = \frac{d}{s}$$



#Q. A man rows to a place 60 km distant and back in a total of 8 hours. He finds that he can row 5 km with the stream in the same time as 3 km against the stream. Find the speed of the man in still water. S_s

16



$$T. \text{ time} = 8 \text{ hrs}$$

$$t_{DS} + t_{up} = 8$$

$$\frac{60}{S_B + S_w} + \frac{60}{S_B - S_w} = 8$$

$$\frac{60 \times 2}{S_w} + \frac{60}{S_w} = 8$$

$$\frac{32}{S_w} = 8$$

$$32 = 8 S_w$$

$$4 = S_w$$

$$\frac{DS}{d=5} \quad \left| \quad \frac{up}{d=3} \right.$$

$$t_{DS} = t_{up}$$

$$\frac{5}{(S_B + S_w)} = \frac{3}{(S_B - S_w)}$$

$$5 S_B - 5 S_w = 3 S_B + 3 S_w$$

$$2 S_B = 8 S_w$$

$$S_B = 4 S_w$$

$$\therefore S_B = 16$$



Escalator:

$p \rightarrow$ speed of person (steps/sec)
 $e \rightarrow$ speed of Escalator (steps/sec)

$$S \times t = d$$



Same
dirⁿ

$$N = pt + et$$

$$\text{Total Steps} = \left(\begin{array}{c} \text{dist} \\ \text{Covered} \\ \text{by} \\ \text{person} \end{array} \right) + \left(\begin{array}{c} \text{dist} \\ \text{Covered} \\ \text{by} \\ \text{Escalator} \end{array} \right)$$

$$\text{Speed} = e + p$$

$$\text{dist} = T \cdot \text{No. of steps (N)}$$

$$\text{Time} = t$$

$$\text{Speed} = \frac{\text{dist}}{\text{time}}$$

$$e + p = \frac{N}{t}$$

$$et + pt = N$$



Escalator:

$$\begin{matrix} s_b > s_w \\ p > e \end{matrix}$$



opposite dirⁿ



$$\text{Speed} = p - e$$

$$\text{time} = t$$

$$\text{T. dist} = N$$

$$\text{Speed} = \frac{\text{dist}}{\text{time}}$$

$$(p - e) = \frac{N}{t}$$

$$pt - et = N$$

$$N = pt - et$$

$$\text{T. Steps} = \left(\begin{matrix} \text{dist} \\ \text{covered} \\ \text{by} \\ \text{person} \end{matrix} \right) - \left(\begin{matrix} \text{dist} \\ \text{covered} \\ \text{by} \\ \text{Escalator} \end{matrix} \right)$$



Escalator :



1. When Person & Escalator are moving in Same direction, then

$$\text{Number of Steps} = \boxed{pt + et}$$

$$\text{Number of Steps} = (\text{steps covered by person}) + (\text{steps covered by escalator})$$

2. When Person & Escalator are moving in Opposite direction, then

$$\text{Number of Steps} = \boxed{pt - et}$$

$$\text{Number of Steps} = (\text{steps covered by person}) - (\text{steps covered by escalator})$$

QUESTION-10

$$M \neq S$$



#Q. Mohan and Sohan climb up or down at the same speed. Mohan takes 8 seconds to go up a certain escalator that is going up, while Sohan takes 12 seconds to go down the same escalator. (If Sohan takes 3 steps for every 2 steps that Mohan takes) how long would Sohan take to go up the same escalator?

t: constant

$$5 \times 1$$

Mohan

$$t = 8 \text{ sec}$$



$$N = mt + et$$

$$N = 2(8) + e(8)$$

$$N = 16 + 8e$$

$$N = 16 + 8(1) = 24$$

Sohan

$$t = 12 \text{ sec}$$



$$N = st - et$$

$$N = 3(12) - e(12)$$

$$N = 36 - 12e$$

$$16 + 8e = 36 - 12e$$

$$20e = 20$$

$$e = 1 \text{ step/sec}$$

Sohan

$$N = st + et$$

$$24 = 3(t) + 1(t)$$

$$24 = 4t$$

$$6 = t$$

A 2.4 sec

B 4 sec

C 4.8 sec

D 6 sec

$$M: 5$$

$$2:3 \rightarrow \text{dist}$$

$$2:3 \rightarrow \text{speed}$$

$$M = 2 \text{ steps/sec}$$

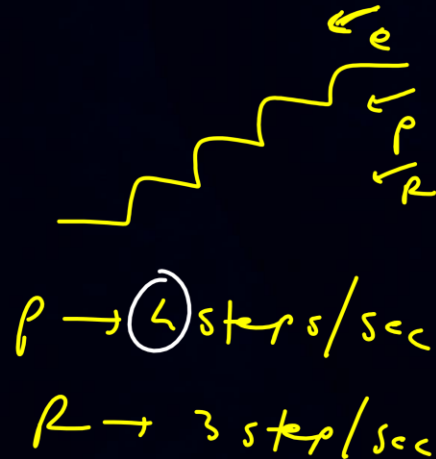
$$S = 3 \text{ steps/sec}$$

QUESTION- 11



#Q. Pushpa and Rocky are on an escalator that moves downwards at a constant speed. Pushpa moves at a speed of 4 steps/second while Rocky moves at a speed of 3 steps/second. When they reach the bottom of the escalator, Pushpa has taken 120 steps and Rocky has taken 108 steps. What is the total number of steps in the escalator visible when it is not moving?

- A** 120
- B** 150
- C** 160
- D** 180 ✓



P

$$Pt = 120$$

$$4t = 120$$

$$t = 30 \text{ sec}$$

$$N = Pt + et$$

$$N = 120 + 30e$$

R

$$Rt = 108$$

$$3t = 108$$

$$t = 36$$

$$N = Rt + et$$

$$N = 108 + 36e \Rightarrow 108 + 36(2)$$

$$\therefore N = 180$$

$$120 + 30e = 108 + 36e$$

$$\frac{120}{12} = \frac{36e}{12}$$

$$10 = 3e$$

$$e = \frac{10}{3}$$



2 Mins Summary

- 1) Concept & Numerical based on 'Problems on Train'
- 2) Concept & Numerical based on 'Boats & Stream'
- 3) Concept & Numerical based on 'Escalator'
- 4) Problems for Practice & CAT PYQs



*Thank
You*